

Inputs

**Patient Info**
Age: 68
Sex: Female
MRN: 98765432

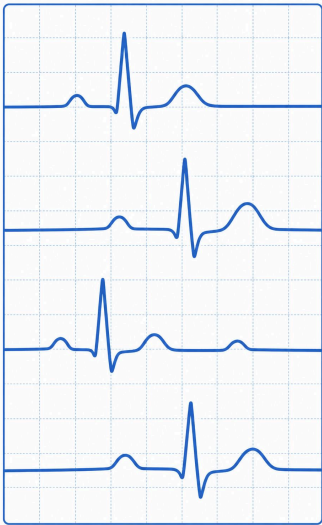
**Demographics**
Race, BMI,

**Labs**
Na, K, HbA1c, Cr, BNP, ...

**Medications**
Metformin, Lisinopril, Atorvastatin,

**Notes**
Progress Note: dyspnea on exertion. Plan:

**Prior Events**
MI (2018), CHF (2021), Hospitalization (2023), ...

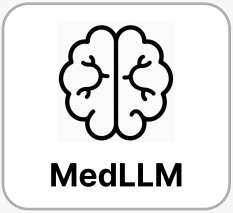


Short Physiological Waveforms

Long EHR Context

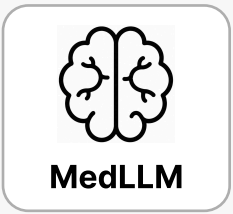
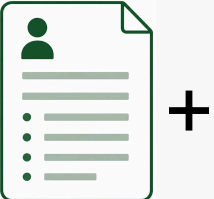
Evaluation Design

EHR Only



Revisit Prediction:
No, the patient will not revisit within 30 days

EHR + Waveform



Revisit Prediction:
Yes, the patient will revisit within 30 days

Conditional Modality Lift

$$\Delta_{\text{lift}} = v(\text{EHR} + \text{Waveform}) - v(\text{EHR})$$

Modality Lift is **Heterogeneous** Across Tasks and Patient Subgroups

Tasks	Subgroups			
	Age	CCI	Sex	Insurance
	Revisit	Positive Lift	Negative Lift	Near Zero Lift
	LOS	Near Zero Lift	Negative Lift	Positive Lift
	Next-Visit	Positive Lift	Near Zero Lift	Near Zero Lift
Mortality	Near Zero Lift	Positive Lift	Negative Lift	Near Zero Lift

 **Positive Lift** (waveform helps)  **Near Zero Lift** (no change)  **Negative Lift** (waveform hurts)

Key Insights



Heterogeneous Lift

Waveform Value Varies Meaningfully Across Subgroups

